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graph TD; A[MIRACL corpus  
2.1M passages] --> B[Arabic tokenizer  
(PyStemmer)]; B --> C[Stopword removal  
(NLTK Arabic, 245 words)]; C --> D[BM25S index build  
 $k_1=0.9$ ,  $b=0.4$ ]; D --> E[Inverted index  
(~5 GB)];
```

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2.1M passages

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